

Erdős Problem 700(i): Complete Solution via a Synchronized Carry Compiler

Nanocodex Erdős 700 project

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Abstract

For

$$f(n) = \min_{2 \leq k \leq \lfloor n/2 \rfloor} \gcd\left(n, \binom{n}{k}\right),$$

we completely solve Part (i) by giving an exact structural characterization of the equality $f(n) = n/q$ for every proper prime-power component q of n . The initial reduction identifies divisibility-minimal carry obstructions. The main structural step then compiles simultaneous realization in all relevant prime bases into one compact finite integer/Boolean system of selectors, prefix products, digits, and borrows. The system has no variable or disjunction for each possible row or multiplier. Taking q to be the largest prime divisor gives the maintained formulation of Erdős Problem 700(i). Taking q to be the greatest exact prime-power component gives the literal formulation in the 1978 Erdős–Szekeres paper, with the necessary prime-power exception. The reductions and both directions of the compiler are checked in Lean; novelty and priority remain matters for independent review.

1 Definitions and statement

Call k *admissible* for n when $2 \leq k \leq \lfloor n/2 \rfloor$. For a prime p , let $C_p(n, k)$ be the number of carries in the base- p addition $k + (n - k) = n$. By Kummer’s theorem,

$$C_p(n, k) = v_p\left(\binom{n}{k}\right).$$

For an admissible k , define its complementary carry weight by

$$W_n(k) = \prod_{p|n} p^{\max\{v_p(n) - C_p(n, k), 0\}}. \quad (1)$$

Fix a positive divisor q of n . Define

$$\text{BoundaryAt}(n, q, d) \iff d \mid n, \quad q < d, \quad \frac{d}{r} \leq q \text{ for every prime } r \mid d. \quad (2)$$

Thus d is a divisibility-minimal divisor of n lying numerically above q . Define

$$\begin{aligned} \text{Realized}(n, d) \iff & \text{there is an } m \geq 1 \text{ such that } dm \leq \lfloor n/2 \rfloor, \\ & C_r(n, dm) \leq v_r(n) - v_r(d) \text{ for every prime } r \mid d. \end{aligned} \quad (3)$$

Finally,

$$\text{BoundarySafeAt}(n, q) \iff \text{no } d \text{ satisfying (2) is realized.} \quad (4)$$

All ranges in (2)–(4) are finite, and the same multiplier m must satisfy every carry constraint in (3).

Theorem 1 (Prime-power threshold theorem). *Let p be prime, let $b \geq 1$, and put $q = p^b$. If $q \mid n$ and $q < n$, then*

$$f(n) = \frac{n}{q} \iff \text{BoundarySafeAt}(n, q).$$

2 The exact complementary weight

Put $G_n(k) = \gcd(n, \binom{n}{k})$. Prime by prime,

$$v_p(G_n(k)) = \min(v_p(n), C_p(n, k)).$$

Since $\min(a, c) + \max(a - c, 0) = a$, (1) gives the exact identity

$$\boxed{G_n(k)W_n(k) = n.} \quad (5)$$

In particular, $W_n(k)$ is a positive divisor of n .

The identity

$$n \binom{n-1}{k-1} = k \binom{n}{k}$$

shows that $n \mid k \binom{n}{k}$. Taking valuations gives

$$v_p(n) - C_p(n, k) \leq v_p(k) \quad \text{whenever the left side is positive,}$$

and hence

$$\boxed{W_n(k) \mid k.} \quad (6)$$

This divisibility is what preserves one common index across all primes.

3 Every proper prime-power threshold is attained

Write $n = cq$. Since $q < n$, we have $c \geq 2$, so $2 \leq q \leq \lfloor n/2 \rfloor$ and q is admissible. The binomial identity

$$\binom{cq}{q} = c \binom{cq-1}{q-1} \quad (7)$$

will give the exact value at this row. In base p , the lowest b digits of both $cq-1$ and $q-1 = p^b-1$ are $p-1$. Repeated application of Lucas's theorem therefore gives

$$\binom{cq-1}{q-1} \equiv 1 \pmod{p}. \quad (8)$$

If $A = \binom{cq-1}{q-1}$, then A is coprime to q , even when $p \mid c$. Consequently

$$\gcd\left(cq, \binom{cq}{q}\right) = \gcd(cq, cA) = c \gcd(q, A) = c = \frac{n}{q}. \quad (9)$$

Combining (5) and (9) yields

$$W_n(q) = q. \quad (10)$$

It follows that

$$f(n) = \frac{n}{q} \iff W_n(k) \leq q \quad \text{for every admissible } k. \quad (11)$$

Indeed, (5) reverses the numerical order between $G_n(k)$ and $W_n(k)$, while (9)–(10) supply equality at an admissible row.

4 The boundary antichain

Lemma 2. *If $W \mid n$, then*

$$q < W \iff \text{some } d \mid W \text{ satisfies } \text{BoundaryAt}(n, q, d).$$

Proof. If $W > q$, choose the numerically least divisor d of W with $d > q$. Then $d \mid n$. For every prime $r \mid d$, the proper divisor d/r also divides W , so minimality gives $d/r \leq q$. Thus d satisfies (2). The reverse implication follows from $q < d \leq W$. $\square$

For $d \mid n$, valuation comparison in (1) gives

$$d \mid W_n(k) \iff C_r(n, k) \leq v_r(n) - v_r(d) \text{ for every prime } r \mid d. \quad (12)$$

If the left side holds, (6) gives $d \mid k$, say $k = dm$ with $m \geq 1$. The admissible endpoint and (12) say exactly that d is realized. Conversely, a witness in (3), together with (12), gives $d \mid W_n(dm)$. Therefore

$$\begin{aligned} W_n(k) \leq q \text{ for every admissible } k \\ \iff \text{no boundary divisor above } q \text{ is realized} \\ \iff \text{BoundarySafeAt}(n, q). \end{aligned} \quad (13)$$

Equations (11) and (13) prove Theorem 1.

5 The compact synchronized compiler

The preceding theorem is already an exact characterization, but its semantic definition quantifies over a shared multiplier. We now describe the stronger form that removes any need to list those multipliers in the statement.

Fix an ordered exact prime factorization

$$n = \prod_{i=1}^r p_i^{a_i}$$

and a baseline B with $2 \leq B \leq n$. A semantic factor tableau selects exponents $0 \leq e_i \leq a_i$ and one positive integer T . Put

$$d = \prod_{i=1}^r p_i^{e_i}, \quad K = dT.$$

The tableau is feasible when

$$B < d, \quad K \leq \lfloor n/2 \rfloor,$$

and, for every active index i with $e_i > 0$,

$$\frac{d}{p_i} \leq B, \quad C_{p_i}(n, K) \leq a_i - e_i. \quad (14)$$

Thus d is a boundary divisor and the one literal row $K = dT$ realizes it. Conversely every realized boundary divisor determines exactly such an exponent vector and shared T . Hence

$$\text{FactorTableau}(n, B) \text{ is feasible} \iff \exists d [\text{BoundaryAt}(n, B, d) \wedge \text{Realized}(n, d)]. \quad (15)$$

The explicit system $\mathcal{G}(F, B)$ is a natural-linear presentation of this tableau. It contains the following finite fields.

1. One-hot Boolean selectors choose each exponent $e_i \in \{0, \dots, a_i\}$.

2. Prefix variables build d and $K = dT$ by multiplying only by the fixed coefficients p_i^e selected in the preceding step.
3. For every i , digit variables expand the same integer K in base p_i through height $h_i = \lfloor \log_{p_i} n \rfloor$.
4. Boolean borrow variables encode subtraction of K from n in each base. Their sum is exactly $C_{p_i}(n, K)$.
5. Boundary and budget rows enforce (14) when $e_i > 0$ and become inactive through checked big- M bounds when $e_i = 0$.

Every multiplication in these rows is by input data or a selected fixed prime power; there is no product of two free integer variables. The system is therefore a finite Presburger, equivalently mixed integer/Boolean, feasibility instance.

Theorem 3 (Compiler soundness and completeness). *For every ordered exact factorization F of n and every $2 \leq B \leq n$,*

$$\boxed{\mathcal{G}(F, B) \iff \text{FactorTableau}(n, B) \text{ is feasible.}}$$

Consequently, for composite $n > 1$,

$$\boxed{f(n) = \frac{n}{P(n)} \iff \neg \mathcal{G}(F, P(n)).} \quad (16)$$

Proof. From a satisfying assignment, one-hotness recovers the exponent vector. The active prefix equalities recover d and $K = dT$ exactly. The digit and borrow rows recover the genuine base- p_i subtraction, so the budget rows give the carry inequalities in (14). This constructs a semantic tableau.

Conversely, from a semantic tableau choose the corresponding one-hot selectors, use the actual prefix products of d and K , take the canonical base- p_i digits of K , and take the genuine borrows in $n - K$. These values satisfy every selector, bound, prefix, boundary, digit, borrow, and budget row. This proves the displayed equivalence. Equation (15), followed by Theorem 1 at $q = P(n)$, gives (16). $\square$

This compilation is materially smaller than enumerating the possible multipliers. The indexed selector, digit, borrow, and prefix coordinates are linear in

$$\sum_i (a_i + 1) + \sum_i (h_i + 1) + r.$$

Since r , $\sum_i a_i$, and every h_i are bounded by the binary length of n , there are $O((\log n)^2)$ indexed coordinates. The fixed coefficients have $O(\log n)$ bits, giving an $O((\log n)^3)$ direct sparse description. This is a compact symbolic characterization, not a claim that the resulting feasibility problem is decidable in polynomial time and not a claim that the equality cases admit a short list of factorization families.

6 The two formulations of Part (i)

Let $P(n)$ be the largest prime divisor of n .

Corollary 4 (Maintained largest-prime formulation). *For every composite $n > 1$ and every ordered exact factorization F of n ,*

$$\begin{aligned} f(n) = \frac{n}{P(n)} &\iff \text{BoundarySafeAt}(n, P(n)) \\ &\iff \neg \mathcal{G}(F, P(n)). \end{aligned}$$

Proof. Apply Theorem 1 with $q = P(n) = P(n)^1$. Since n is composite, $P(n)$ is a proper divisor of n . The second equivalence is Theorem 3. $\square$

The 1978 paper instead uses

$$Q(n) = \max_{p|n} p^{v_p(n)}, \quad (17)$$

the greatest exact prime-power component of n .

Corollary 5 (Literal 1978 formulation). *For every composite $n > 1$ and every ordered exact factorization F of n ,*

$$\begin{aligned} f(n) = \frac{n}{Q(n)} &\iff n \text{ is not a prime power and } \text{BoundarySafeAt}(n, Q(n)) \\ &\iff n \text{ is not a prime power and } \neg \mathcal{G}(F, Q(n)). \end{aligned}$$

Proof. If n is not a prime power, an exact component attaining (17) is a proper prime-power divisor, so Theorem 1 applies. If $n = p^a$ is composite, then $Q(n) = n$. Lucas’s theorem shows that every interior binomial coefficient is divisible by p , while Kummer’s theorem at $k = p^{a-1}$ gives valuation exactly one. Thus $f(p^a) = p > 1 = n/Q(n)$, so the equality is impossible. The second equivalence follows from Theorem 3 at baseline $Q(n)$. $\square$

Remark 6. The two formulations are genuinely different. For $n = 12$, the gcd values for $k = 2, 3, 4, 5, 6$ are 6, 4, 3, 12, 12, so $f(12) = 3$. Here $P(12) = 3$ and $Q(12) = 4$: the historical equality $f(12) = 12/Q(12)$ holds, whereas the maintained equality $f(12) = 12/P(12)$ does not.

7 Formal verification and source trail

The parameterized theorem is `Erdos700PartI.f_eq_div_primePow_iff_boundarySafeAt`. The maintained and historical corollaries are respectively `Erdos700PartI.f_eq_div_iff_boundarySafe` and `Erdos700PartI.erdos_700_i_historical`. They are exposed in `lean-part1/PartIVerify.lean`. The repository verifier builds the promoted roots, rejects placeholders, audits dependencies, and runs finite regressions. Lean reports only the standard Mathlib axioms `propext`, `Classical.choice`, and `Quot.sound`; it reports no `sorryAx` dependency.

The semantic bridge is

`Erdos700PartI.factorTableauFeasible_iff_exists_boundary_realized`, and the explicit compiler theorem is `Erdos700PartI.ExplicitG.explicitG_iff_factorTableauFeasible`. Lean checks both directions, including one-hot decoding, prefix products, canonical digit and borrow construction, inactive-row bounds, and the exact carry-budget identity. The primary historical source is P. Erdős and G. Szekeres, “Some number theoretic problems on binomial coefficients,” *Australian Mathematical Society Gazette* 5 (1978), 97–99, archival PDF.